Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_18

1. Summary of Key Deviations

The analysis of Run_18 reveals a complex signature with two distinct, simultaneous fault patterns: a critical sensor failure and a separate physical issue of fuel over-delivery.

- **p_FMU (Critical):** The pressure sensor at the Fuel Metering Unit is the site of the most severe fault. It shows a **major negative drift** beginning in state s2, worsening to a **complete signal dropout (-99.4%)** in state s5. This is a definitive sensor failure.
- **Flow_Pump (Moderate):** In parallel, the `Flow_Pump` signal shows a significant positive **offset**, reading **+12.2%** higher than the healthy baseline.
- **Flow_Bypass (Moderate):** Corresponding to the high pump flow, the `Flow_Bypass` signal also shows a large positive **offset** of up to **+21.4%**, as the excess flow is diverted.
- **Flow_Engine1-4 (Minor):** The engine flows are all slightly positive (up to **+6.7%**), indicating some of the excess fuel is reaching the combustion chamber.
- **Other Signals (Nominal):** Other pressures and `Flow_PRV` remain nominal, showing the system is managing the physical fault without over-pressurizing.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**p_FMU Sensor & Main Fuel Pump**	**Dual Fault: Sensor Dropout & Pump Over-performance**	This is the only hypothesis that explains all symptoms. There are two independent failures: 1) The **p_FMU sensor has failed**, causing the dropout. 2) The **Main Fuel Pump is significantly over-performing** (+12.2%), causing the high readings on `Flow_Pump`, `Flow_Bypass`, and `Flow_Engine`. These two faults are happening concurrently.

2	**p_FMU Sensor**	**Sensor Dropout / Total Failure**	This explains the critical -99.4% reading. However, a sensor-only fault **cannot** explain the separate, physically consistent pattern of over-performance seen in `Flow_Pump`, `Flow_Bypass`, and the engine flows.
3	**Main Fuel Pump**	**Significant Over-performance**	This explains the high flow readings across the board. However, it does **not** explain why the `p_FMU` pressure sensor, specifically, drops to zero while other pressure sensors remain healthy.
4	**Catastrophic Leak at FMU**	**Major Leak in FMU Casing/Piping**	A massive leak *at* the FMU could theoretically cause the local pressure (`p_FMU`) to drop to zero. However, this contradicts all other data: a leak would cause flow and pressure to *decrease* downstream, but here `Flow_Engine` is high, and `Flow_Pump` is also high.
5	**Multiple Sensor Failures**	**p_FMU Dropout & Other Sensor Errors**	It is extremely unlikely that the `p_FMU` sensor would fail (dropout) at the same time as the `Flow_Pump` and `Flow_Bypass` sensors all fail with a positive offset in a way that perfectly mimics a real physical over-performance scenario. The dual physical/sensor fault is far more plausible.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Total Failure of the p_FMU Pressure Sensor AND (2) a concurrent, separate instance of Main Fuel Pump Over-performance.**

Reasoning: The data clearly shows two distinct problems that are not physically dependent on each other. The first is the undeniable dropout of the `p_FMU` sensor, which is an isolated electrical or component failure. The second is a classic fuel over-delivery signature, where the pump is producing too much flow, and the rest of the system is reacting accordingly (high bypass flow, high engine flow). Neither of these faults can cause the other. A pump performing at +12% would not cause an adjacent pressure sensor to read -99%. Likewise, a failed pressure sensor would not cause the physical pump to deliver more fuel. Therefore, the only logical conclusion is that the system has suffered two simultaneous, independent failures.